Tutorial Lecture 10

Topics: Inclusion/Exclusion & Approximation

Question 1. *Inclusion/Exclusion Warm-up*

How many integers in the range $[1, 2, \ldots, 10000]$ are either divisible by 5, 11 or 19?

Question 2. *Counting Perfect Matchings*

A *perfect matching* in a graph $G = (V, E)$ is a subset of the edges $M \subseteq E$ such that every vertex $v \in V$ is incident to exactly one edge in M. Even though it is possible to find a perfect matching a graph G if it exists in polynomial time, the problem of counting the number of perfect matchings in a graph G is hard (it is $\#P$-complete).

We will construct an algorithm for this problem that uses $\mathcal{O}^*(2^n)$ time and polynomial space. We note that faster algorithms exist.

1. Show that the number of perfect matchings in a graph G equals the number of edge subsets $M \subseteq E$ of size $n/2$ with the property that every vertex $v \in V$ is incident to *at least one* edge in M.
2. Show that for a given vertex subset $U \subseteq V$, the number of edge subsets $F \subseteq E$ of size $n/2$ in $G[U]$ can be computed in polynomial time (there exist a direct formula for this number). Note that the edge subsets counted in this subquestion are *not* perfect matchings.
3. Show how to use (b) and (c) and the principle of inclusion/exclusion to construct $\mathcal{O}^*(2^n)$ time and polynomial space algorithm for counting perfect matchings.

Open Pool Questions

Lecture 10: Inclusion/Exclusion & Approximation

Question 3. *Inclusion/exclusion for set cover*

Let U be any set of size m ($m = |U|$), and let $\mathcal{F}$ be a set of subsets of U. A set cover $\mathcal{C} \subset \mathcal{F}$ is a set of sets that together cover all sets in U, i.e., $\bigcup_{S \in \mathcal{C}} S = U$. A minimum set cover is a set cover using the smallest number of sets from $\mathcal{F}$ as possible.

Give an $\mathcal{O}^*(2^m)$ time and polynomial-space algorithm that counts the number of minimum set covers of $\mathcal{F}$.

Question 4. *Approximation non-symmetric TSP*

Suppose we have an instance of TRAVELLING SALESMAN PROBLEM that is not symmetric but still fulfils the triangle inequality. I.e., batery usage by an electric car in an mountainous area. We know that for every two cities i and j, we have that $d(i, j) \leq 2d(j, i)$, and of course, $d(j, i) \leq 2d(i, j)$.

Give a c-approximation algorithm for this problem that runs in polynomial time.

Homework Lecture 10

Hand-In: Tutorial 11, Tuesday March 17th

Please prepare the following homework question for the Hand-In date. Please bring your homework on paper. You get one point for handing in the homework and you get another point for participating in a round of peer-feedback. Taking place during the tutorial.

Question 5.

We consider the VERTEX COVER PROBLEM. In this exercise, we look to another approximation algorithm for this problem. This algorithm uses Depth-First Search.

The algorithm works as follows. Suppose G is connected. Run the Depth First Search algorithm, and obtain the DFS-tree T. Now, let C be the set of all vertices that have at least one child in T, i.e., C is the set of all vertices that are not a leaf in T. Output C.

1. Prove that C is a vertex cover of G.
2. Give a proof for the following fact. Let T be a tree with r vertices that are not a leaf and ℓ leaves. Then there is a matching in T with at least $r/2$ edges. (Hint: use induction.)
3. Show that G has a matching of size at least $|C|/2$.
4. Prove that the algorithm outputs a vertex cover whose size is at most two times the size of a minimum vertex cover.
5. The algorithm above was stated for connected graphs. What would you do for graphs that are not connected?